

Somatic Polygenic Risk Scoring: A Novel Framework for Metastasis Risk Stratification

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April 21, 2026

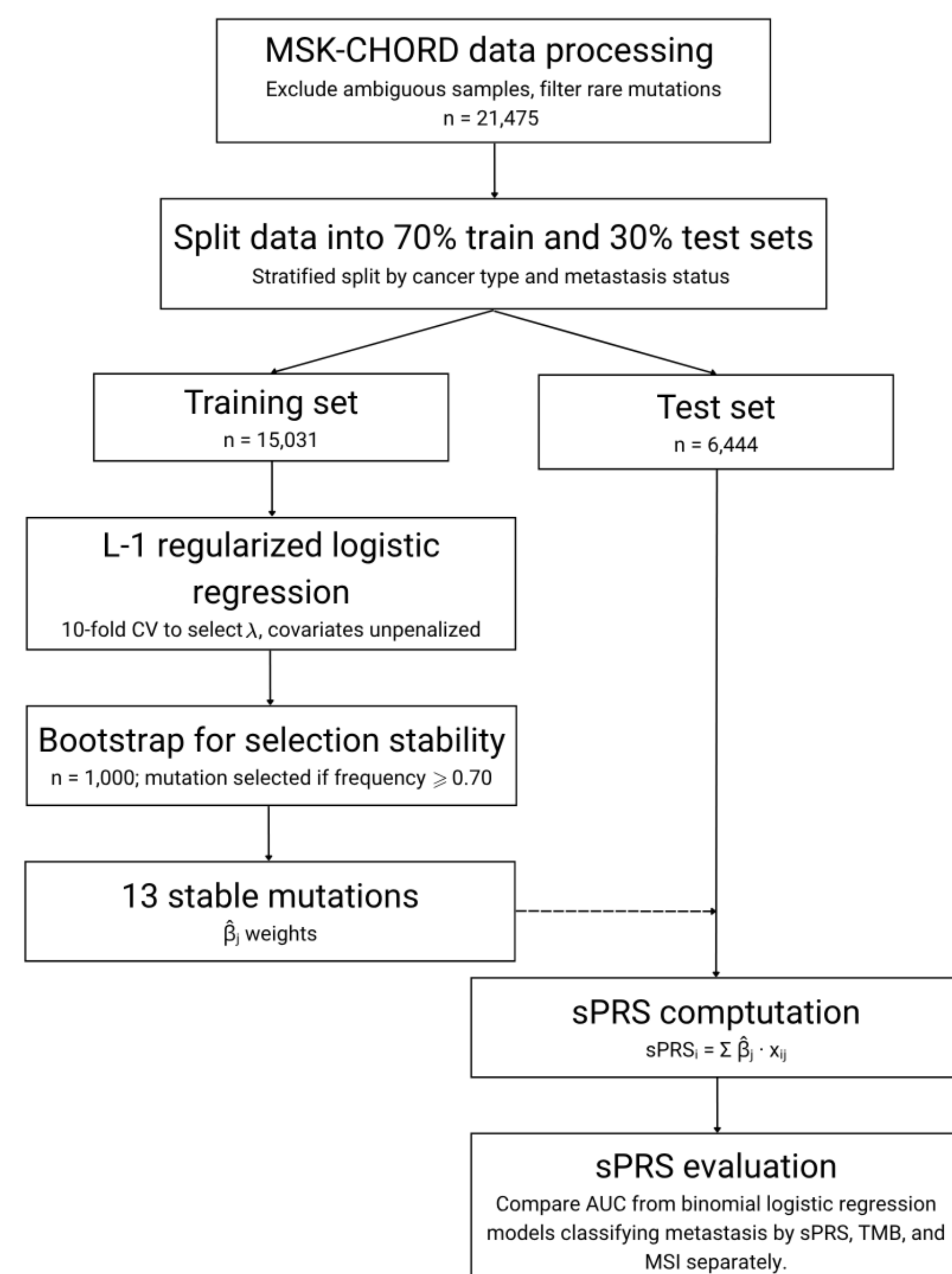
INTRODUCTION

- Metastasis is a major cause of cancer mortality.
- Current genomic biomarkers (tumor mutational burden- TMB, microsatellite instability- MSI) are single-statistic summaries primarily used to inform treatment selections.
- Germline polygenic risk scores (PRS) aggregate many small genomic effects into one score indicating genetic risk of a biological event.

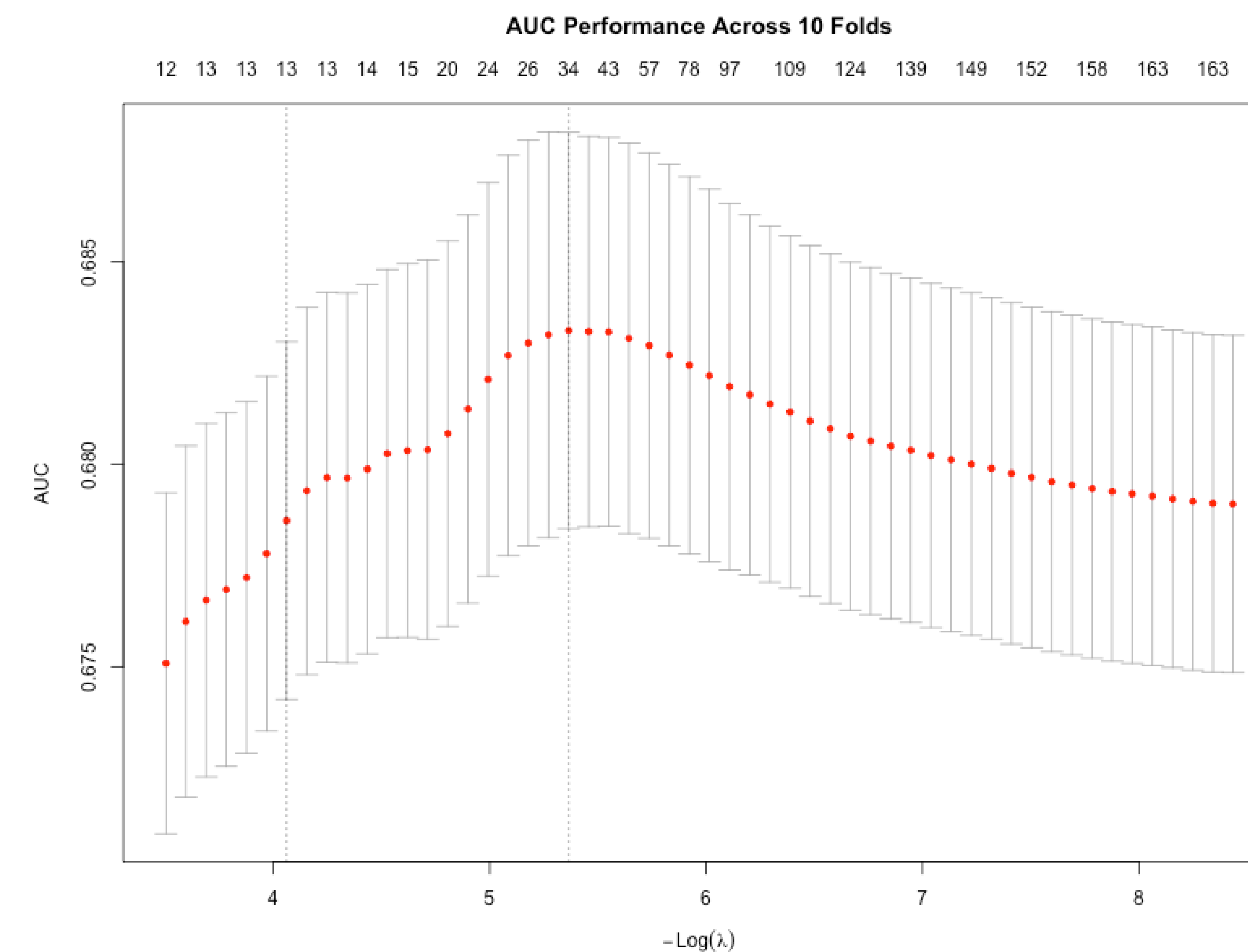
PURPOSE

Derive a somatic polygenic risk score (sPRS) from tumor genomes for metastasis risk stratification.

METHODS



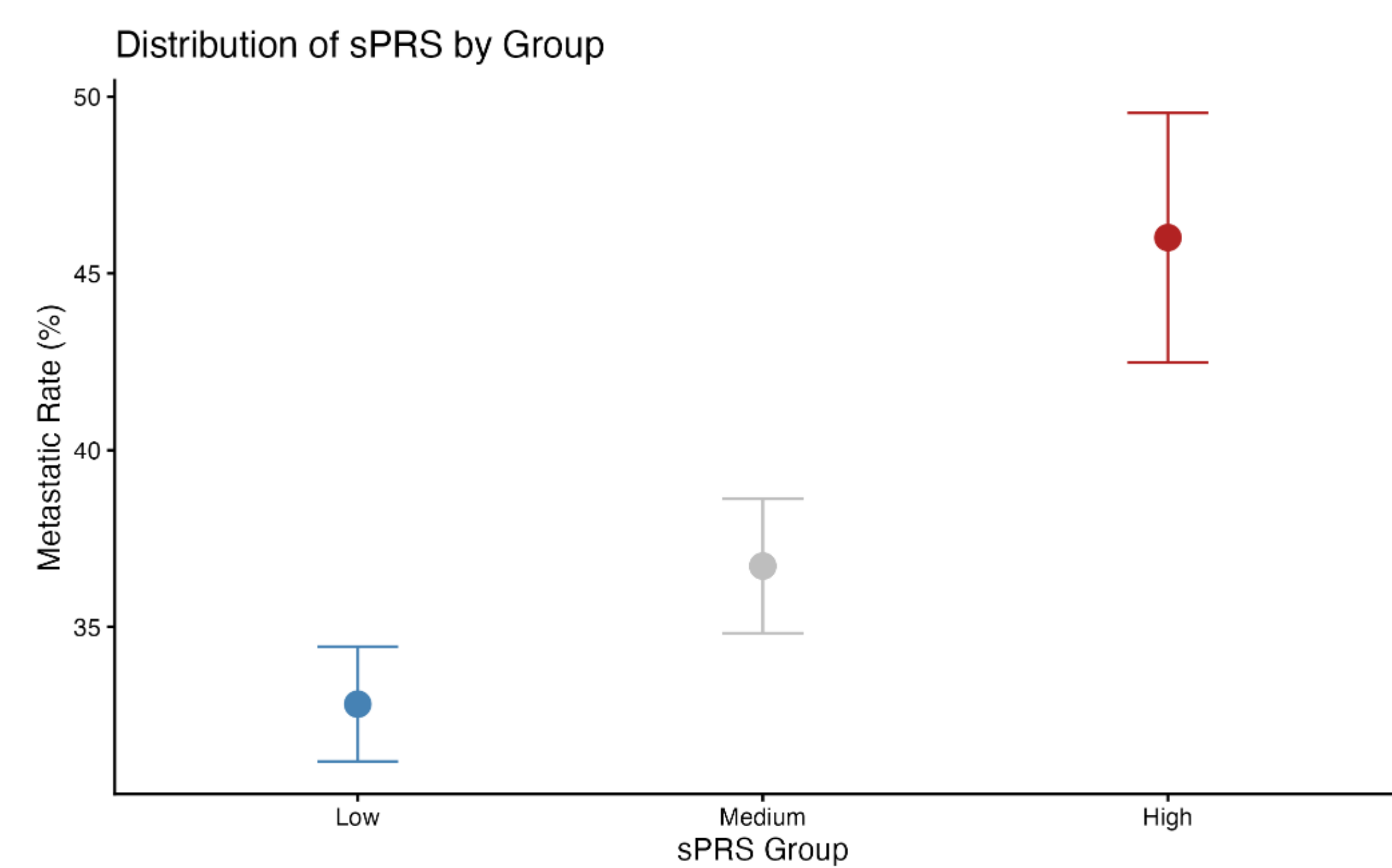
RESULTS



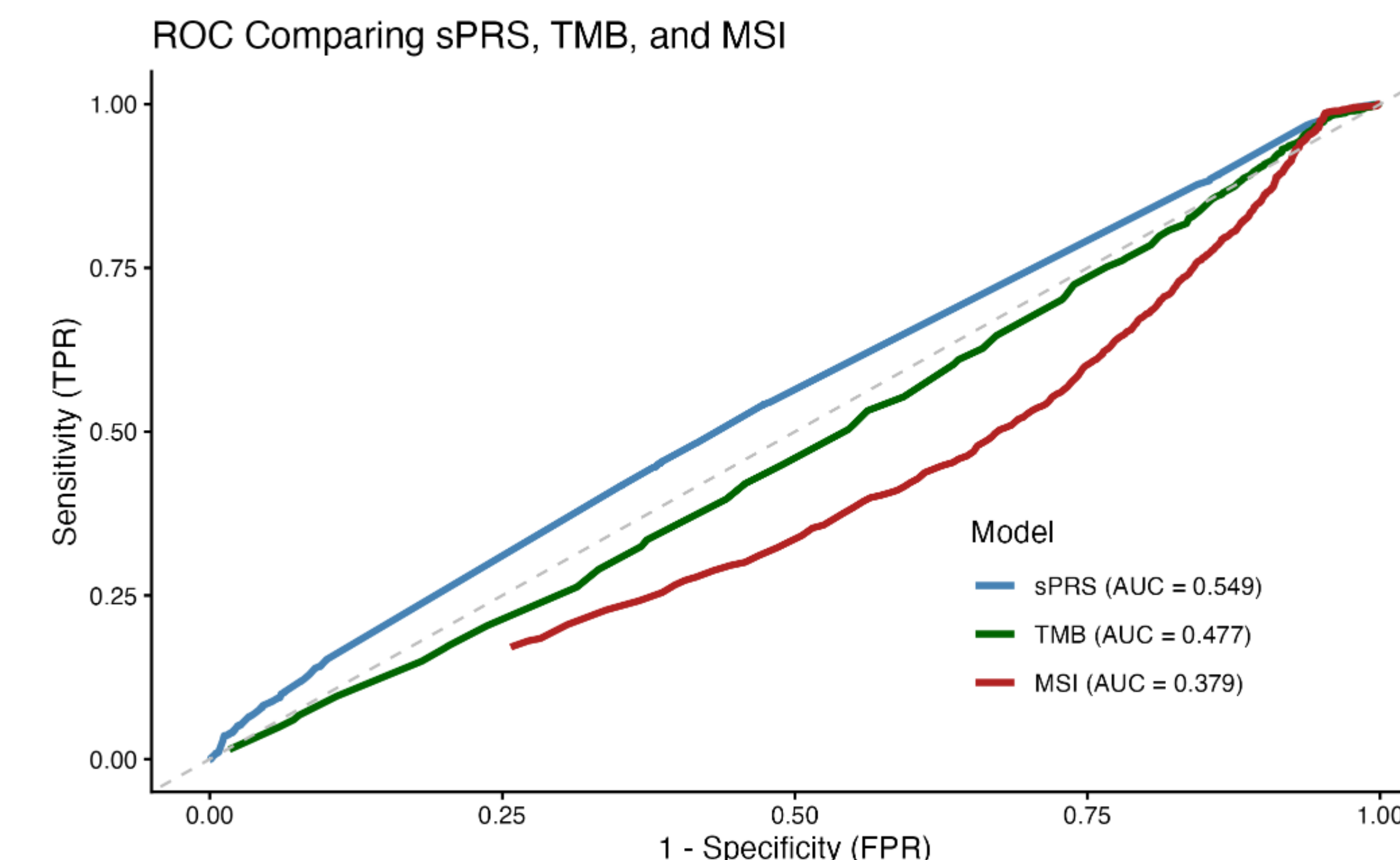
12 unpenalized covariates and 22 mutations preserved at the minimum lambda. 12 unpenalized covariates and 1 mutation preserved at 1-SE lambda. Minimum lambda chosen for clinical importance.

Mutations Retained in sPRS	
Gene	Mutation Type
AR	Missense
ARID1A	Nonsense
ARID1B	Missense
ERBB4	Missense
ESR1	Missense
FBXW7	Missense
KEAP1	Missense
KMT2C	Missense
PIK3CA	Missense
RNF43	Frame Shift Deletion
TP53	Missense
TP53	Nonsense
TP53	Splice Site

Of 22 mutations, 13 remained after bootstrapping with an inclusion frequency cutoff of 0.70.



Overall chi-squared test was significant ($p < 0.001$). All pairwise comparisons (Low vs. Medium, Low vs. High, Medium vs. High) were statistically significant after Bonferroni correction ($\alpha = 0.017$).



sPRS shows a higher AUC than traditional genomic biomarker measures such as TMB and MSI when discriminating between metastatic and primary patients.

DISCUSSION

- The sPRS showed modest but significant discrimination between metastatic and primary tumors (AUC = 0.54).
- sPRS outperformed TMB (AUC = 0.477) and MSI (AUC = 0.379) as genomic predictors of metastatic status.
- Significant association between sPRS group (Low, Average, and High) and metastatic rate suggests the aggregation of metastatic burden captured meaningful contributions beyond individual genomic biomarkers.
- Note the low AUC, which can be explained by the biological complexity of metastasis that extends beyond somatic mutations.

CONCLUSION

- This study proposes a novel framework for aggregating small effect somatic mutations into a single metastasis risk score (similar to a germline PRS).
- sPRS is associated with metastatic status and outperforms traditional genomic biomarker scores such as TMB and MSI.
- A higher sPRS is associated with a significantly higher metastatic rate, providing clinical interpretability for the constructed sPRS.
- However, cross-sectional tumor sequencing limits causal inference, and future work should validate the sPRS framework in longitudinal cohorts with matched primary-metastatic pairs.
- An additional further direction is validating this study on a larger cohort in a more general population. MSK-CHORD is a very specific subset of cancer patients, which is not representative of the cancer population as a whole.